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Sandbox for Agentic Application with Cloud Native Stack

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Agent tools need a stronger boundary



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An agent can write code, invoke tools, install packages, and follow network paths.

Untrusted input · The prompt is not a trust boundary.

Useful credentials · The workload may reach data, tokens, and internal services.

Long-lived loops · A failed task can leave processes, storage, network access, and cost behind.

Execution needs an explicit security and lifecycle boundary.

A sandbox is a lifecycle—not a Pod



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Create → configure → execute → observe → reset → destroy

Before start · Bind identity, policy, resources, runtime, and image inputs.

During execution · Expose only the filesystem, network, credentials, and tools the task requires.

After execution · Collect evidence, remove resources, revoke access, and release capacity—even on failure.

A Pod is one object inside this lifecycle; it is not the lifecycle itself.



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Strong isolation needs a separate kernel



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SHARED-KERNEL CONTAINER

Fast startup and high density

Mature OCI and Kubernetes workflow

The host kernel remains the shared trust boundary

VM-BACKED SANDBOX

A guest kernel per sandbox boundary

Kata preserves the container workflow

Startup and memory overhead must be engineered,
not ignored



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Compose, do not reinvent

Control plane · runtime · images · lifecycle

One stack, two planes, one security boundary



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Agentic application

Kubernetes Agent Sandbox API

Kubernetes scheduling · RuntimeClass · CRI

containerd · Nydus snapshotter

Kata runtime-rs · Dragonball VMM

KVM / optional PVM · guest kernel boundary

IMAGE PATH

OCI registry



Dragonfly distribution



Nydus lazy image

Control stays declarative.
Execution crosses a VM boundary.



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Kubernetes stays the control plane



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DESIRED STATE

Agent Sandbox · Create and manage the sandbox lifecycle declaratively.

Scheduler · Use labels, taints, tolerations, and a dedicated sandbox pool.

RuntimeClass · Select the xolis-kata handler explicitly.

NODE CONTRACT

CRI · Keep the standard Kubernetes-to-containerd contract.

containerd · Map the handler to Kata and the image path to Nydus.

Policy · Prevent ordinary workloads from bypassing the sandbox runtime.



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Kata defines the boundary; PVM widens reach



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KATA + DRAGONBALL

VM-isolated execution with the container workflow
runtime-rs with an embedded VMM path
Guest kernel and root filesystem per sandbox

KVM + OPTIONAL PVM

AWS first path: native nested KVM on supported
nodes
PVM is an experimental fallback when VT is not
exposed
Separate AML, node pool, RuntimeClass, and
observability



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Nydus and Dragonfly move less data, sooner



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NYDUS

- OCI-compatible image artifacts
- Lazy, on-demand filesystem reads
- containerd snapshotter integration

DRAGONFLY

- Distribution, cache, and peer-to-peer delivery
- Reduce repeated registry traffic at scale
- Add after the minimal execution path is reproducible



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Disposability requires guaranteed cleanup

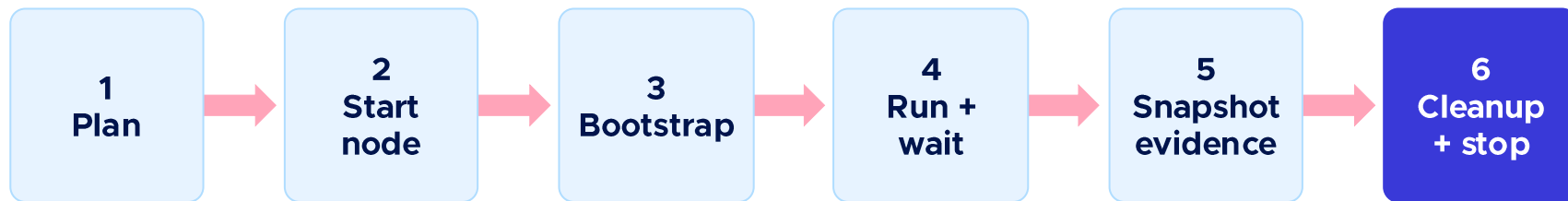


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Success and failure converge on the same cleanup path

The current snapshot is diagnostic resource evidence—not yet a VM checkpoint.



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Xolis separates durable and disposable work



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DURABLE INFRASTRUCTURE

OpenTofu · VPC, EKS, IAM, system nodes, and sandbox nodes.

State · Remote and locked before shared use.

Change control · Explicit plan, apply, and destroy.

DISPOSABLE TEST CYCLE

Python CLI · Start, bootstrap, run, observe, delete, and stop.

Evidence · Command log and Kubernetes resource snapshot.

Safety · Cleanup and node stop execute in a finally path.



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Demo: validate, benchmark, clean up



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1 Validate the complete service

```
python3 tools/xolis_aws_lab.py --config tools/xolis_aws_lab.json service run
```

2 Compare cold and warm claims

```
python3 tools/xolis_aws_lab.py --config tools/xolis_aws_lab.json benchmark run --iterations 3
```

Measured once in Tokyo • Ready: 25.5 s cold → 1.4 s warm

Directional only • One sample per mode is not an SLA.

Watch: node → service → claim → command → cleanup → scale-to-zero

The MVP works; now measure it



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VALIDATED NOW

Agent Sandbox v0.5.3 service

Kata + Dragonball on EKS M8i

Acceptance and cleanup automation

Ready: 25.5 s cold; 1.4 s warm (n=1)

NEXT, IN ORDER

20–30 runs; report p50 / p95 / p99

Compare Nydus with stable OCI

Add OIDC, quotas, audit, and ingress

Stream output; then automate capacity



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A decade of upstream work converges here



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The contribution is integration discipline—not another private runtime stack.



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Build upstream first; measure every boundary



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DESIGN PRINCIPLES

- Upstream projects before private forks
- Replaceable layers with standard contracts
- Dedicated runtime pools with explicit policy
- Cleanup is part of correctness

CONTRIBUTE THE MISSING PROOF

- Run repeated cold and warm baselines
- Validate Nydus against stable OCI
- Measure reliability, density, and cost
- Publish results and fixes upstream



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Secure agents by composing proven cloud native boundaries

github.com/gnawux/xolis · Q&A